

DISCRETE STRUCTURES · REVIEW

# Sets, Proofs, Functions & Relations

*Review — Part 2*

$\forall$   $\exists$   $\Rightarrow$   $\Leftrightarrow$   $\in$   $\subseteq$   $\cup$   $\cap$   $\times$   $\equiv$

# Four pillars

1

## Cartesian Products

Ordered pairs and the set  $A \times B$ .

2

## Proofs

Theorems, counterexamples, and proof methods.

3

## Functions

Injective, surjective, bijective, predicates.

4

## Relations

Binary relations and equivalence relations.

# Cartesian Product of Two Sets

- Given sets  $A$  and  $B$ , their Cartesian product is written  $A \times B$ .
- It contains every ordered pair whose first element comes from  $A$  and whose second element comes from  $B$ .
- Order matters — the pair  $\langle a, b \rangle$  is ordered.

$$A \times B = \{ \langle a, b \rangle : a \in A \text{ and } b \in B \}$$

# $A \times B$ — a worked example

$$A = \{1, 2, 3\} \quad B = \{\triangle, \star\}$$

$$A \times B = \{(1, \triangle), (1, \star), (2, \triangle), (2, \star), (3, \triangle), (3, \star)\}$$

**Order matters**

$$B \times A \neq A \times B$$

*...unless  $A = B$ .*

*$A \times B$  has  $|A| \cdot |B| = 6$  ordered pairs*

# Theorems and Proofs

## Theorem

A true mathematical statement.

## Proof

A convincing logical argument that the statement is true.

**Example** If  $x, y$  are rational then  $x \cdot y$  is rational.

Let's prove this statement.

# Disproving with a Counterexample

- A universal statement (“for all ...”) is disproved by a single counterexample.

## Can disprove

- “All swans are white.”
- One non-white swan is enough to make it false.

## Existential claims cannot be disproved this way

- “There exists at least one green swan.” — an existential claim: no single case refutes it.

## Is the following statement true?

- “All unicorns are red.”
- vacuously true: there are no unicorns to contradict it.

# Counterexample in action

**Claim:** If  $x \in \mathbb{N} \wedge (x \bmod 2 = 0)$ , then  $x$  is a power of 2.

*Disprove by counterexample*

1

Let  $x = 6$ .

2

$6 \in \mathbb{N}$  and  $6 \bmod 2 = 0$  ✓

3

but 6 is not a power of 2. ✗

4

**$\therefore$  the claim does not hold.**

# Implications and “If and Only If”

$$p \Rightarrow q$$

*If  $p$  is true, then  $q$  must be true.*

$$p \Leftrightarrow q$$

*$p$  and  $q$  imply each other.*

## Truth is formal

“If the moon is made of cheese, then  $2 + 2 = 7$ .” A material implication can be true even when its parts are not.

$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$$

*A biconditional is two implications, both directions.*

# Four Proof Methods

1

## Direct proof

Argue straight from the hypotheses to the conclusion.

2

## Proof by contradiction

Assume the negation; derive an impossibility.

3

## Proof by induction

Base case, then step from  $n$  to  $n + 1$ .

4

## Proof by construction

Exhibit an object with the required property.

**Note:** *proof by construction*  $\neq$  *proof by example*.

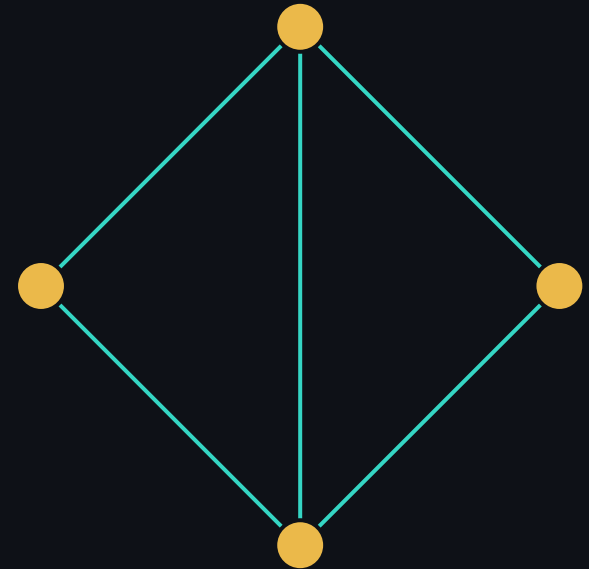
# Direct Proof — the Handshake Lemma

- For a graph  $G = (V, E)$ , let  $\deg(v)$  be the number of edges incident to  $v$ .

$$\sum_{v \in V} \deg(v) \text{ is even}$$

## *Proof.*

- Counting degrees, each edge  $\{u, v\}$  is counted exactly twice — once in  $\deg(u)$  and once in  $\deg(v)$ .
- So  $\sum \deg(v) = 2 \cdot |E|$ , which is even. ■



each edge  $\rightarrow +2$  to the degree sum

# Set Identity — Strategy

$$(A \cup B)^c = A^c \cap B^c$$

To prove two sets are equal, show mutual inclusion:

**1** Show  $X \subseteq Y$

$$(A \cup B)^c \subseteq A^c \cap B^c$$

**2** Show  $Y \subseteq X$

$$A^c \cap B^c \subseteq (A \cup B)^c$$

# Element Chase — Forward Inclusion

Part ①: show  $(A \cup B)^c \subseteq A^c \cap B^c$

Let  $x \in (A \cup B)^c$

$\Rightarrow x \notin A \cup B$

$\Rightarrow x \notin A$  and  $x \notin B$

$\Rightarrow x \in A^c$  and  $x \in B^c$

$\Rightarrow x \in A^c \cap B^c$  ✓

## Part ②

The reverse inclusion holds by reversing each step — every  $\Rightarrow$  above is in fact an  $\Leftrightarrow$ .

# Relations and Functions

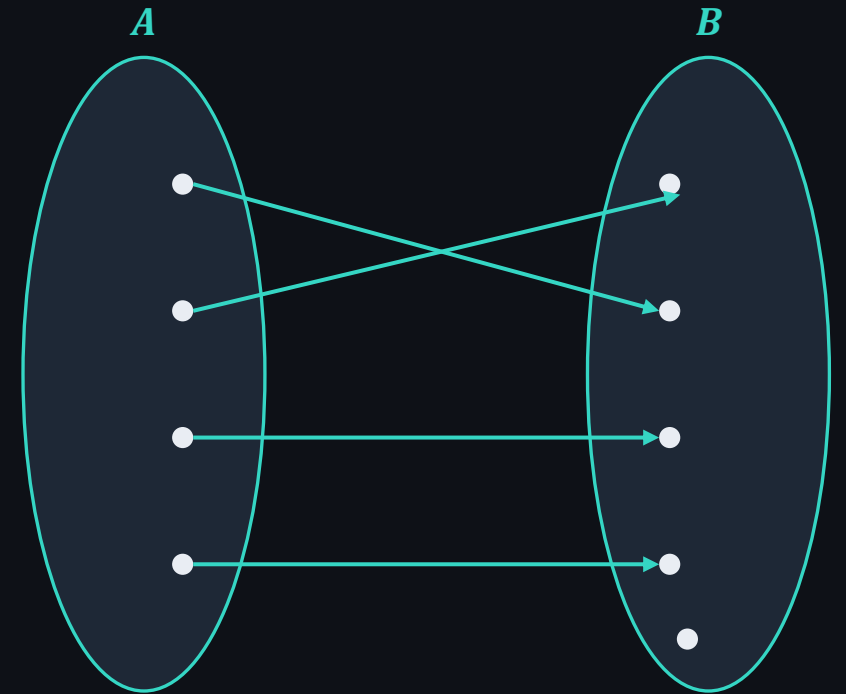
- A function is a mapping between two sets, governed by a rule.

$$f: A \rightarrow B$$

For every  $a \in A$ :

- $f(a)$  must be defined
- $f(a)$  must be unique
- $f(a) \in B$

$A$  = domain     $B$  = codomain     $R$  = range,  $R \subseteq B$



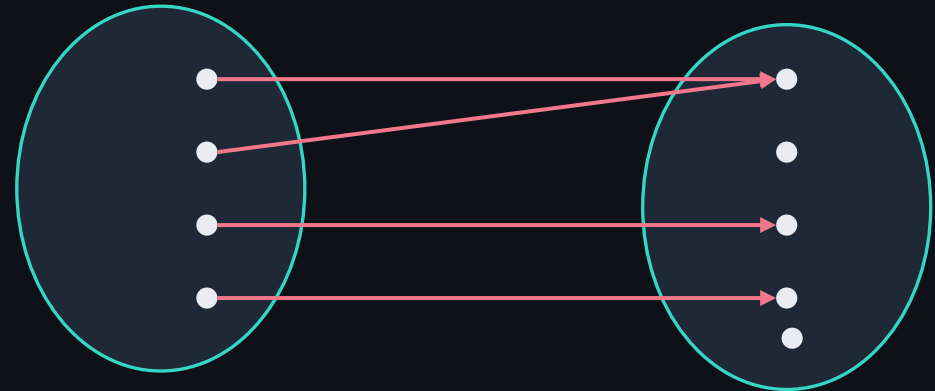
# Injective (One-to-One)

- Distinct inputs always give distinct outputs.

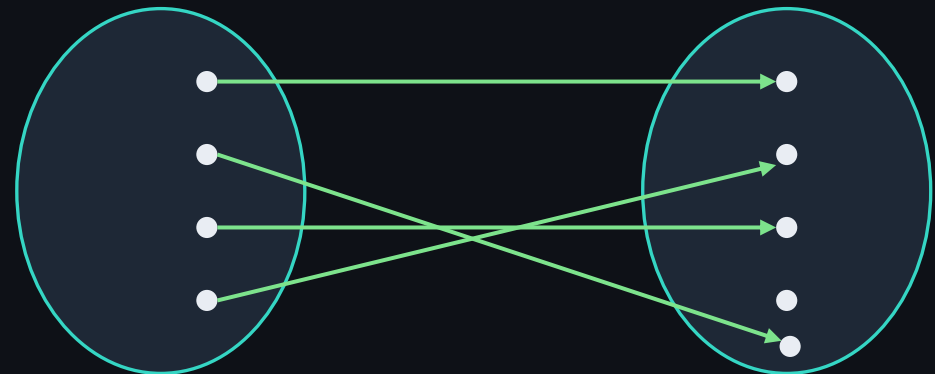
$$\forall a_1, a_2 \in A : a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)$$

*No two arrows may land on the same target.*

Not one-to-one



One-to-one

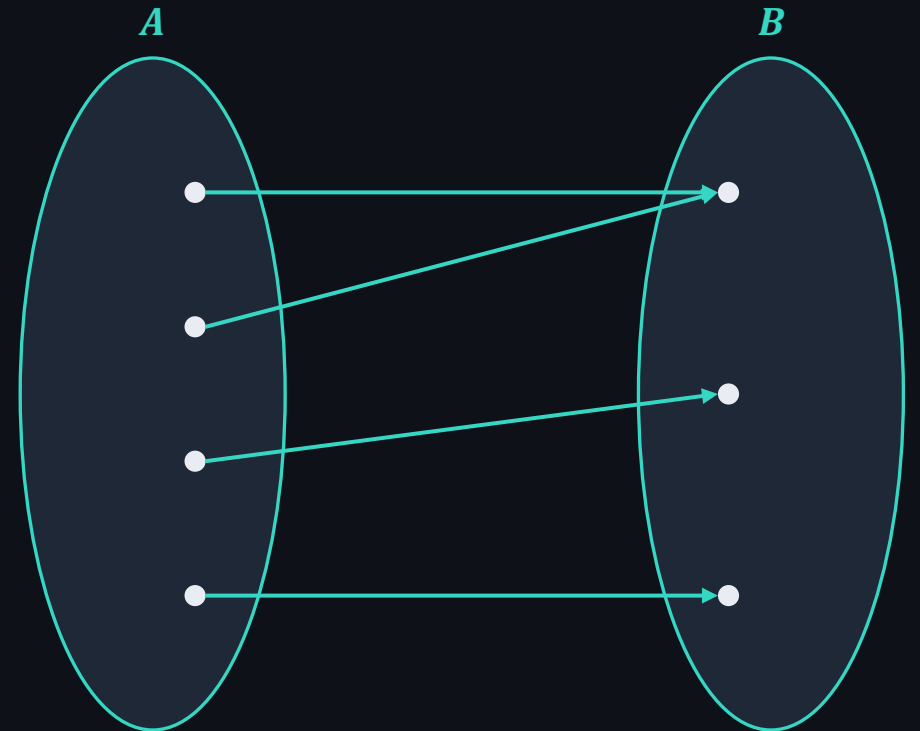


# Surjective (Onto)

- Every element of the codomain is hit by some input.
- Equivalently, the range equals the codomain:  $R = B$ .

$$\forall b \in B: \exists a \in A: f(a) = b$$

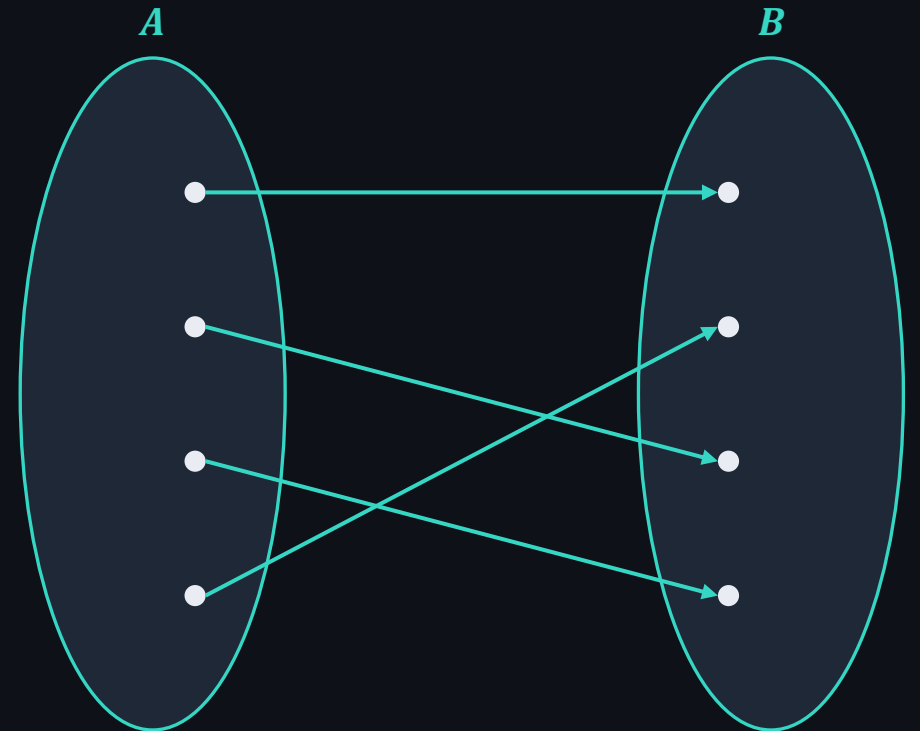
*No target is left without an incoming arrow.*



# Bijection

- A function that is both injective and surjective.
- It pairs each element of  $A$  with exactly one element of  $B$ , and vice versa — so it is invertible.

**bijection** = injective + surjective



# Predicates

- A predicate (or property) is a function whose range is  $\{ \text{True}, \text{False} \}$ .

**isEven** :  $\mathbb{Z} \rightarrow \{ \text{True}, \text{False} \}$

**isEven**(x) = {  
    **True**, if x is even  
    **False**, if x is odd

*range*

**True**

**False**

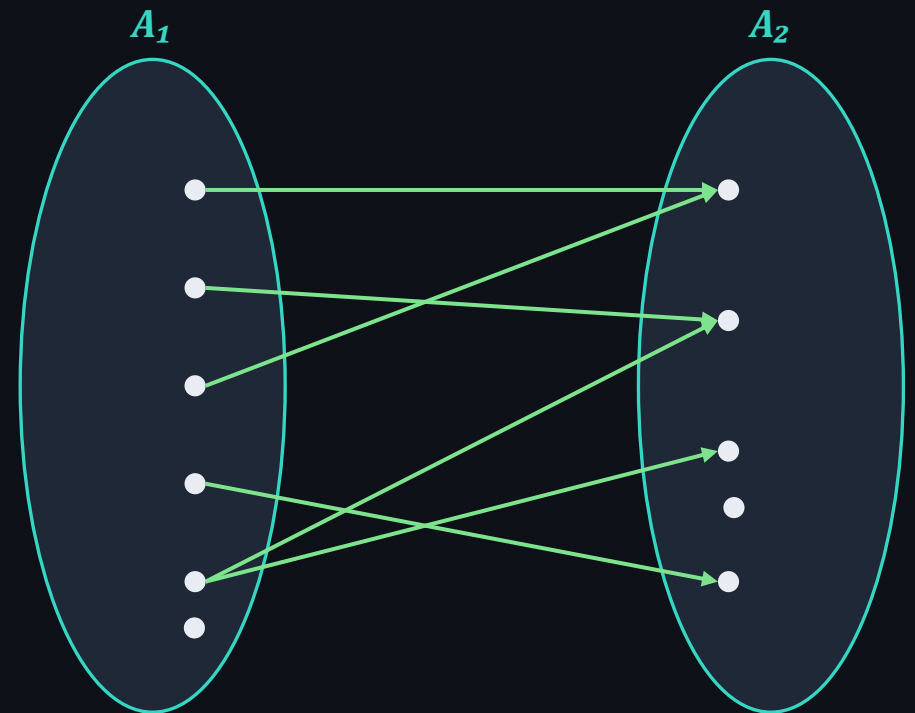
# Relations — the General Idea

- A  $k$ -ary relation  $R$  on sets  $A_1, A_2, \dots, A_k$  is any subset of their Cartesian product.

$$R \subseteq A_1 \times A_2 \times A_3 \times \dots \times A_k$$

- A relation simply selects which tuples “belong together.”
- Example ( $k = 2$ ):  $R \subseteq A_1 \times A_2$  picks out a subset of all pairs.

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# Binary Relations

- In practice we usually take  $k = 2$  — these are binary relations.

## Between two sets

$$R \subseteq A \times B$$

Relates elements of one set to another.

## On a single set

$$R \subseteq A \times A$$

Let's us define reflexivity, symmetry, transitivity — the ingredients of an equivalence relation.

# Equivalence Relations

A binary relation  $R \subseteq A \times A$  is an equivalence relation when it is:

1

**Reflexive**

$$\forall x \in A : x R x$$

2

**Symmetric**

$$\forall x, y \in A : x R y \Rightarrow y R x$$

3

**Transitive**

$$\forall x, y, z \in A : (x R y \wedge y R z) \Rightarrow x R z$$